Avi Kumar Singh

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Summary

Software Engineer with 4 years shipping production systems in Python and Java. Currently serving vision-language models (Qwen, Gemma) with vLLM, developing video curation capabilities with a leading AI partner.

Technical Skills

AI/ML: vLLM, LLM/VLM inference, Qwen, Gemma, CLIP, PyTorch, Hugging Face Transformers
Video Processing: FFmpeg, PyAV, NVIDIA DALI
Languages: Python, Java, JavaScript, SQL
Frameworks: Flask, Spring Boot, React
Cloud, Data & DevOps: AWS, Azure, Docker, MySQL, DynamoDB, Redis, Jenkins, Git, Linux

Experience

V-Nova September 2024 – Present
Software Engineer *London, UK*

- Sped up a partner's video curation pipeline 2.3x via visual token pruning and proprietary-format transcoding.
- Served Qwen and Gemma VLMs with vLLM on 8x A100 GPUs (AWS) for video curation.
- Led development of a Python video-processing orchestrator, scaling on demand from 10 to ∼150 AWS nodes.
- Reduced page load time from minutes to seconds with lazy loading and Redis caching.

Espire Infolabs May 2023 – September 2024
Software Engineer *London, UK*

- Engineered a dynamic document engine with Spring Boot and React, reducing licensing costs.
- Automated CCM migration with a Java app on Azure Functions, cutting manual effort by 70%.

Amazon August 2022 – February 2023
Software Engineer *Cambridge, UK*

- Maintained a high-availability Java microservice powering Alexa Linked Data queries, onboarding new use cases with full unit test coverage.
- Migrated a dependency service using shadow-mode traffic, reducing fault rates.

Education

National Institute of Technology Trichy July 2018 – July 2022
Bachelor of Technology in Computer Science, Minor in Economics

Internship

Oracle June 2021 – July 2021
Intern – Application Engineer *Bengaluru, India*

- Built Capacity Manager, a Spring and Oracle Cloud tool that reduced pod provisioning failures by up to 5%, with a React dashboard for real-time cloud insights.
- Automated cloud alerts to admins with a Java cron job integrated with Slack.

DailyObjects December 2019 – January 2020
Intern – Android Engineer *New Delhi, India*

- Built new Android UI views in Kotlin and Jetpack using TDD for an app with 500,000+ users; automated testing with Firebase Test Lab and CircleCI.

Projects

Quick Mail CLI | *Python, Docker, PyPI* | *GitHub* | *PyPI* Spring 2021

- Developed and dockerised command line interface to send email quickly. Averages 35 installs every month.